

READING PASSAGE 2

Black Holes and Event Horizons

A The study of black holes and event horizons has undergone a remarkable transformation over the past several decades. What was once a niche area of academic inquiry has developed into a rich and diverse field of research, drawing on insights from a wide range of disciplines and theoretical traditions.

B Historically, scholarly understanding of this topic was limited by both theoretical frameworks and available methodologies. Early researchers operated within relatively narrow conceptual boundaries, and their findings, while groundbreaking for their time, often failed to capture the full complexity of the phenomena they sought to explain. The development of new research tools and analytical techniques in the latter half of the twentieth century fundamentally transformed the landscape, enabling scholars to investigate questions that had previously been considered beyond the reach of empirical inquiry.

C Contemporary research has identified several critical factors that determine outcomes in this domain. The interplay between environmental conditions, technological capabilities, and social organisation creates a complex adaptive system in which small changes can produce disproportionately large effects. Understanding these dynamics requires analytical approaches that move beyond simple cause-and-effect reasoning to embrace the principles of complexity science and systems thinking.

D The empirical record is further strengthened by a series of meta-analyses that have synthesised findings from hundreds of individual studies. These comprehensive reviews have consistently identified moderate to large effect sizes, with the most robust findings demonstrating remarkable consistency across different research designs, measurement approaches, and analytical strategies. This convergence of evidence from multiple independent lines of inquiry considerably strengthens confidence in the validity of the core theoretical propositions.

E The practical application of research findings has proven more challenging than many initially anticipated. The translation of academic knowledge into actionable policy recommendations requires navigating a complex landscape of political constraints, economic interests, and social dynamics that academic models often simplify or overlook. Implementation failures in several high-profile initiatives have underscored the gap between theoretical understanding and practical capacity, prompting calls for more engaged and context-sensitive approaches to research.

F Looking ahead, the field stands at a critical juncture. Emerging technologies offer unprecedented opportunities for data collection and analysis, potentially enabling researchers to address questions that have long remained beyond the reach of empirical investigation. At the same time, the accelerating pace of social and technological change means that the phenomena under study are themselves evolving, requiring theoretical frameworks that can accommodate dynamism and discontinuity rather than assuming stability and equilibrium.

G In conclusion, the study of black holes and event horizons represents a vital and dynamic area of contemporary research. The progress that has been made in understanding the key mechanisms and dynamics provides a solid foundation for both further scholarly inquiry and practical application. The challenges that remain should be seen not as reasons for discouragement but as opportunities for continued intellectual growth and social contribution.

Question 1

Choose the correct letter, A, B, C or D.

1 Which of the following best describes the author's view on future developments?

- A. economic incentives and institutional constraints
- B. individual personality traits
- C. random chance
- D. government policies alone

Question 2

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

2 The author notes that researchers have called for more _____ approaches to investigation.

Question 3

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

3 Early researchers had access to the same technologies as modern scholars.

Question 4

Choose the correct letter, A, B, C or D.

4 What finding emerged from the landmark study discussed in the passage?

- A. economic incentives and institutional constraints
- B. individual personality traits
- C. random chance
- D. government policies alone

Question 5

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

5 All critics agree that the research findings are universally applicable.

Question 6

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

6 Researchers used data from more than _____ countries in the landmark study.

Question 7

Choose the correct letter, A, B, C or D.

- 7 What does the passage identify as a significant challenge to current understanding?
- A. weak effects across studies
 - B. moderate to large effect sizes with consistency
 - C. contradictory findings
 - D. results only in specific populations

Question 8

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

- 8 Meta-analyses have confirmed the reliability of the core findings in this field.

Question 9

Choose the correct letter, A, B, C or D.

- 9 The author suggests that early studies in this field were limited by:
- A. weak effects across studies
 - B. moderate to large effect sizes with consistency
 - C. contradictory findings
 - D. results only in specific populations

Question 10

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

- 10 The author believes that all research questions in this field have been answered.

Question 11

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

- 11 Dynamic _____ processes can amplify or dampen initial effects in complex systems.

Question 12

Choose the correct letter, A, B, C or D.

- 12 What does the passage identify as a significant challenge to current understanding?
- A. weak effects across studies
 - B. moderate to large effect sizes with consistency
 - C. contradictory findings

D. results only in specific populations

Question 13

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

13 Case studies have documented the effects of _____ change on the system.